

Experiment 10: Feature Extraction using DenseNet121 and Classification with SVM

Theory:

DenseNet121 is a pre-trained convolutional neural network model that has shown strong performance in image recognition tasks. By removing the classification head (`include_top=False`), we can use DenseNet as a feature extractor. This technique is commonly used in transfer learning, where features from a model trained on a large dataset (e.g., ImageNet) are used for a new, smaller dataset. In this experiment, DenseNet121 is used to extract deep features from CIFAR-10 images, which are then classified using a linear Support Vector Machine (SVM). This approach benefits from the representational power of deep CNNs and the classification capability of SVMs, especially when training data is limited.

Algorithm:

1. Load CIFAR-10 dataset and select a small subset for quick experimentation.
2. Preprocess the images to match DenseNet input requirements using ImageNet scaling.
3. Load DenseNet121 with pre-trained weights and exclude the classification layer (`include_top=False`).
4. Use DenseNet121 to extract feature vectors from the training and test images.
5. Train a Support Vector Classifier (SVC) on the extracted features and corresponding labels.
6. Evaluate the classifier using the test data, and plot the confusion matrix.
7. Simulate training with increasing data sizes to observe changes in accuracy.

Flowchart (written format):

- Load CIFAR-10 data
- Preprocess with ImageNet standard
- Initialize DenseNet121 (no top layer)
- Extract features using DenseNet
- Train SVM on features
- Predict and evaluate accuracy
- Plot confusion matrix and training curve